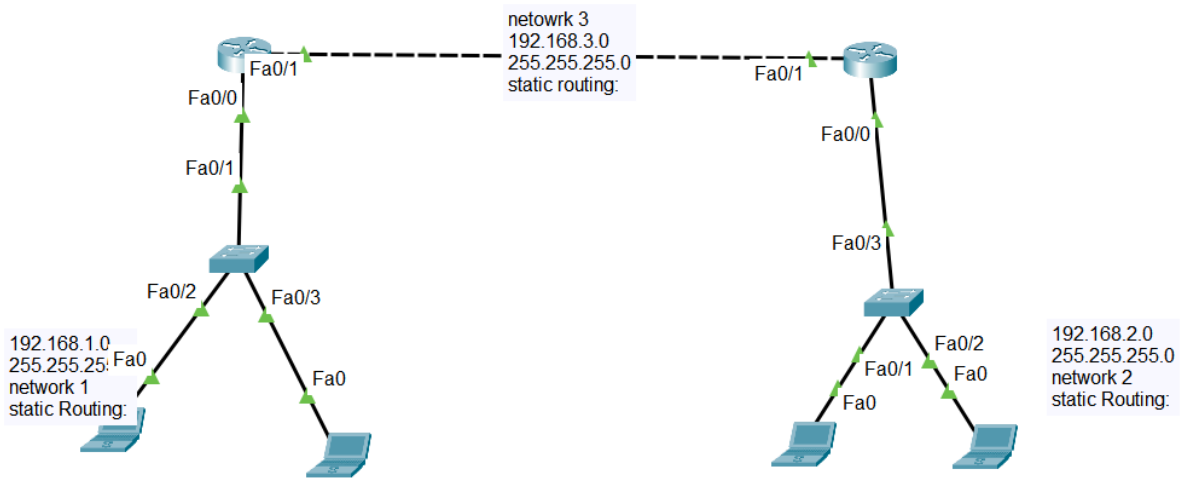
# NETWORKING PROJECT REPORT

Implementation of DHCP and Static Routing
Using Cisco Packet Tracer




**CCNA Practical Project - 01**

**Prepared by:** _____

**Software:** Cisco Packet Tracer

**Year:** 2026

# 1. Introduction

This project demonstrates the implementation of Dynamic Host Configuration Protocol (DHCP) and static routing using Cisco Packet Tracer. The topology contains two separate local area networks (LANs), each connected to its own router through a switch. The two routers are interconnected through a third IPv4 network, 192.168.3.0/24.

DHCP is configured on each router to automatically provide IPv4 configuration to the laptops in its local LAN. Static routes are configured on both routers so that devices in Network 1 can communicate with devices in Network 2, subject to successful configuration and testing.

# 2. Project Objectives

• Design two LANs connected through two Cisco routers.

• Configure IPv4 addresses on router interfaces.

• Configure DHCP pools on both routers for dynamic laptop addressing.

• Configure static routes to reach the remote LAN.

• Verify interface status, DHCP leases, routing entries and end-to-end connectivity.

# 3. Tools and Network Devices

| Component | Quantity | Purpose |
|---|---|---|
| Cisco routers | 2 | Provide routing and DHCP services |
| Switches | 2 | Connect laptops within each LAN |
| Laptops | 4 | End devices used for connectivity tests |
| Cisco Packet Tracer | 1 software | Network design, configuration and simulation |

# 4. Network Topology

The topology is divided into three IPv4 networks. Network 1 is the left-side LAN, Network 2 is the right-side LAN, and Network 3 is the transit network between Router1 and Router2.

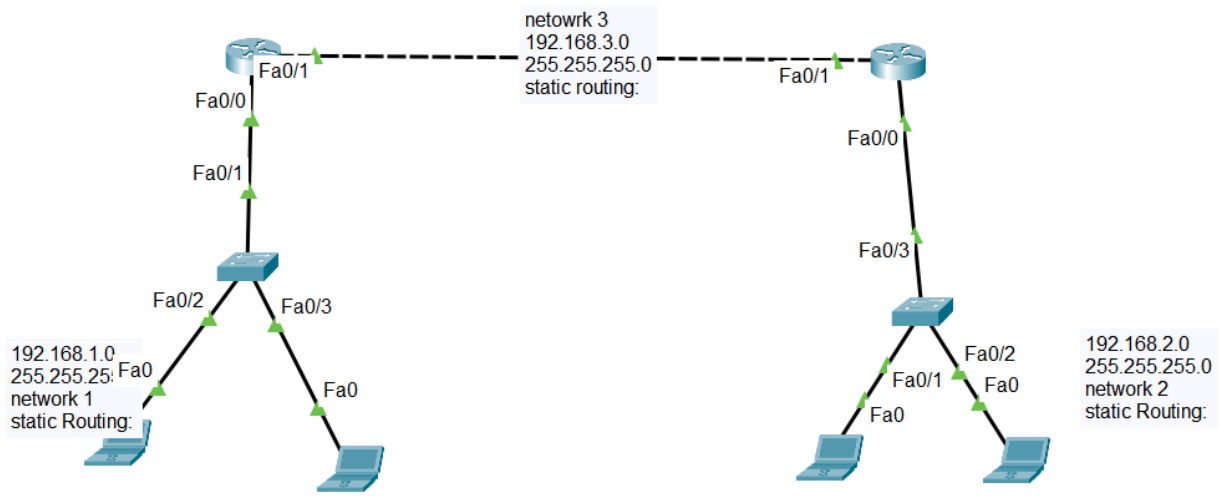



Figure 1. Topology image supplied for this project. Confirm that the saved Packet Tracer file matches this diagram.

# 5. IP Addressing Scheme

All three networks use a /24 subnet mask (255.255.255.0). The router interface addresses and gateways are based on the configuration details provided for this project.

| Network | Network address | Subnet mask | Purpose |
|---|---|---|---|
| Network 1 | 192.168.1.0/24 | 255.255.255.0 | LAN 1 |
| Network 2 | 192.168.2.0/24 | 255.255.255.0 | LAN 2 |
| Network 3 | 192.168.3.0/24 | 255.255.255.0 | Router transit link |

| Device | Interface | IP address | Subnet mask |
|---|---|---|---|
| Router1 | Fa0/0 | 192.168.1.1 | 255.255.255.0 |
| Router1 | Fa0/1 | 192.168.3.1 | 255.255.255.0 |
| Router2 | Fa0/0 | 192.168.2.1 | 255.255.255.0 |
| Router2 | Fa0/1 | 192.168.3.2 | 255.255.255.0 |

### Laptop addressing

Laptop1 and Laptop2 are configured to obtain their addresses through the DHCP pool on Router1. Laptop3 and Laptop4 are configured to obtain their addresses through the DHCP pool on Router2. Their exact leased addresses must be copied from each laptop's Desktop → IP Configuration screen after DHCP succeeds; those lease values were not supplied when this report was prepared.

# 6. DHCP Configuration

**DHCP** stands for Dynamic Host Configuration Protocol. It automatically assigns client devices an IP address and other network settings, such as the default gateway. In this project, each router serves its directly connected LAN.

### Router1 DHCP pool (Network 1)

```
enable
configure terminal
ip dhcp pool cisco1
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
exit
```

### Router2 DHCP pool (Network 2)

```
enable
configure terminal
ip dhcp pool cisco2
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
exit
```

**Syntax correction:** Cisco IOS uses `default-router` inside a DHCP pool, not `default-route`. The configuration above uses the correct command.

To check DHCP leases, use `show ip dhcp binding`. To check pool information, use `show ip dhcp pool`. The laptops must be set to DHCP in Desktop → IP Configuration.

# 7. Static Routing Configuration

Static routing is a method in which an administrator manually enters routes into a router's routing table. Each router needs a route to the other router's LAN. Network 3 is directly connected to both routers, so it appears as a connected network and does not need an additional static route.

### Router1 route to Network 2

```
ip route 192.168.2.0 255.255.255.0 192.168.3.2
```

### Router2 route to Network 1

```
ip route 192.168.1.0 255.255.255.0 192.168.3.1
```

### Static route command format

```
ip route [destination-network] [subnet-mask] [next-hop-IP]
```

On Router1, the next hop is Router2's transit-interface address (192.168.3.2). On Router2, the next hop is Router1's transit-interface address (192.168.3.1).

# 8. Verification and Testing

| Command / test | Purpose |
|---|---|
| show ip interface brief | Check router interface IP addresses and operational status |
| show ip route | Inspect connected and static routes |
| show ip dhcp pool | Inspect DHCP pool information |
| show ip dhcp binding | View IP addresses leased to clients |
| show running-config | Review the active router configuration |
| ping <remote-laptop-IP> | Test connectivity to a laptop in the other LAN |

Record the actual results after running the tests. Do not mark a test as successful unless it has been performed and the result is visible. Suggested evidence includes interface status, DHCP bindings on both routers, routing tables on both routers, laptop IP Configuration screens, and a successful ping between laptops in different LANs.

### Test results

| Test | Result |
|---|---|
| Laptop1 and Laptop2 obtain DHCP addresses | To be verified |
| Laptop3 and Laptop4 obtain DHCP addresses | To be verified |
| Router1 and Router2 transit interfaces are up | To be verified |
| Static routes appear in both routing tables | To be verified |
| Laptop in Network 1 can ping laptop in Network 2 | To be verified |

# 9. Conclusion

This project provides practical experience in IPv4 addressing, DHCP configuration, router interface configuration and static routing. Two LANs are connected through a third transit network. DHCP is intended to assign IP settings automatically to the laptops, while static routes provide paths between the two LANs. Successful operation should be confirmed using the verification commands and end-to-end ping tests documented above.

**Documentation note:** This report is prepared from the topology and configuration details supplied. The actual DHCP lease addresses and test outcomes remain to be filled in after checking the running Packet Tracer simulation.